

```
import numpy as np

weights, bias = np.array([0.4, 0.6]), -0.2

def linear_combination(input: np.ndarray) -> np.ndarray:
    Z = np.dot(input, weights) + bias
    A = relu(Z)
    return Z, A

def relu(n: np.ndarray) -> np.ndarray:
    return np.maximum(0, n)
```

✓ Task 1

```
input = np.array([2.0, -1.0])

z, y = linear_combination(input)
print(f"Linear Combination \t {z}")
print(f"ReLU \t\t\t {y}")
```

Linear Combination	5.551115123125783e-17
ReLU	5.551115123125783e-17

✓ Taks 2

```
inputs = np.array([-2.0, 1.0])

z,y = linear_combination(inputs)
print(f"Linear Combination \t {z}")
print(f"ReLU \t\t\t {y}")
```

Linear Combination	-0.4000000000000001
ReLU	0.0